Matplotlib Cheatsheet

Import

```python
import matplotlib.pyplot as plt
import numpy as np
```

Display the Plot

```python
day = np.array([1, 2, 2, 4, 5, 6, 7])
coffee = np.array([1, 2, 1, 1, 1, 0, 1])

plt.figure()           # Set up canvas
plt.plot(day, coffee)  # Create the plot
plt.show()             # Display the plot 🪄
```

Title & Axis Labels

```python
plt.title('Coffee Drank This Week')
plt.xlabel('Day of the Week')
plt.ylabel('Cups of Coffee')
```

Multiple Lines & Legends

```python
plt.plot(time, uber, label='Uber prices')
plt.plot(time, lyft, label='Lyft prices')
plt.legend()
```

Line Colors and Line Styles

```python
plt.plot(x, y, color='red', linestyle='--')
```

Pandas DataFrame / Import CSV

```python
import pandas as pd

df = pd.DataFrame({
  'day': [1, 2, 3, 4, 5],
  'temperature': [54, 59, 48, 51, 59]
})

df = pd.read_csv('weather.csv')

plt.plot(df['day'], df['temperature'])
```

Scatter Plot

```python
plt.scatter(df['hours'], df['scores'])
```

Bar Graph

```python
plt.bar(df['major'], df['num_students'])
```

Pie Charts

```python
plt.pie(votes, labels=flavors)
```

Object-Oriented Style

```python
fig, ax = plt.subplots()

ax.plot((df['Month'], df['Sales'])
ax.plot(df['Month'], df['Profit'])
ax.set_title('My Plot')

plt.show()
```

Subplots

```python
# Single row or column (axes is 1D array)
fig, axes = plt.subplots(1, 3)

axes[0].plot(x, y)          # left
axes[1].plot(x, y)          # middle
axes[2].plot(x, y)          # right

# Multiple rows & columns (axes is 2D array)
fig, axes = plt.subplots(2, 3)

axes[0, 0].plot(x, y)       # top-left
axes[0, 2].plot(x, y)       # top-right
axes[1, 1].plot(x, y)       # bottom-middle
```

Markers

```python
ax.plot(df['Mon'], df['Sales'], marker='o')
ax.plot(df['Mon'], df['Profit'], marker='s')
```

Codédex